Automation & QA Developer Assessment

Brief documentation of work completed | Jaswant

Task 1 — Web Application Testing and Bug Analysis

Tested the React Redux RealWorld Conduit application locally using manual UI exploration, browser network inspection, source-code review, and Jest test-file discovery. Five issues were documented with reproduction steps, expected and actual results, severity, and suspected causes. Findings included an external stylesheet loading failure, API request failures, missing required-field validation in forms, and tag-input validation gaps.

Root cause investigated: When an API request failed, the Redux promise middleware accessed `error.response.body` without verifying that `error.response` existed. When no response object was available, this caused a secondary TypeError. The recommended fix is to guard response/body access and dispatch a consistent fallback error object. The external API returned HTTP 530 during testing, limiting completion of some end-to-end flows; this limitation was recorded in the QA report.

Task 2 — GitHub Morning Brief Automation (n8n)

Built an n8n workflow that searches GitHub repositories tagged with `artificial-intelligence`, sorts them by stars, and selects the top five. An IF node checks whether each repository has more than 1,000 stars. For qualifying items, a second GitHub API request retrieves README metadata. A Merge node combines repository and README information, and a Code node formats the final records.

Output: The workflow appends repository name, description, star count, language, repository URL, README URL, README size, README status, error detail, priority, and generation timestamp to a Google Sheet named *GitHub Morning Brief*.

Error handling and testing: The README HTTP Request node is configured to continue using regular output on failure. This was tested with a deliberately invalid README path; the workflow continued rather than stopping at that request. The endpoint was restored and a successful execution processed five records in approximately 6.07 seconds.

Tools and Deliverables

Task 1: React Redux RealWorld application, browser DevTools, source-code review, and QA report PDF.
Task 2: n8n, GitHub REST API, JavaScript Code nodes, IF, Merge, and Google Sheets using OAuth2 credentials.
Evidence: Public GitHub repository containing the QA report, exported workflow JSON, workflow canvas screenshot, successful execution screenshot, and README.

Note: The n8n workflow has an hourly Schedule Trigger configured but was left inactive after testing to avoid repeated automatic appends to the Google Sheet.